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LIST OF ABBREVIATIONS

CRS – Crop Recommendation System

STGCN – Spatio-Temporal Graph Convolutional Network

GNN – Graph Neural Network

IDW – Inverse Distance Weighting

RMSE – Root Mean Square Error

MAE – Mean Absolute Error

R^2 – Coefficient of Determination

DHM – Department of Hydrology and Meteorology

T2M – Temperature at 2 meters

RH2M – Relative Humidity

PRECTOT – Precipitation

PS – Surface Pressure

WS10M – Wind Speed

CHAPTER 1

INTRODUCTION

1.1 Background

Agriculture remains the backbone of Nepal's economy, particularly in rural and mountainous regions. Crop selection decisions depend strongly on local weather conditions such as temperature, humidity, and rainfall. In recent years, machine learning-based Crop Recommendation Systems (CRS) have been developed to assist farmers in making data-driven decisions. However, these systems rely heavily on the accuracy of underlying meteorological forecasts.

In mountainous terrain, weather variability is highly complex. Elevation differences, valley inversions, orographic lifting, and rain shadow effects create microclimates that cannot be captured accurately by simple spatial interpolation methods. Most spatial forecasting models assume that geographically close locations exhibit similar climatic behavior. This assumption is derived from Tobler's First Law of Geography, which has traditionally informed spatial interpolation and modeling approaches [1], [2].

While this principle works reasonably well in flat landscapes, it fails in high-relief environments such as the Himalayas.

Recent advancements in Spatio-Temporal Graph Neural Networks (STGNNs), built upon Graph Convolutional Network (GCN) theory [3], particularly STGCN [4], have shown strong forecasting performance. However, most implementations construct graph adjacency matrices using geographic distance. In mountainous regions, geographic proximity may not reflect atmospheric similarity.

This study proposes a Correlation-Adaptive Weather Graph framework in which graph topology is derived from historical statistical similarity between stations rather than physical distance.

1.2 Statement of the Problem

The current standard approach for graph-based weather forecasting uses distance-based adjacency matrices defined as:

$$A_{ij}^{dist} = \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right)$$

Where d_{ij} represents the geodesic distance between stations i and j , and σ is a scaling parameter controlling spatial influence.

This formulation assumes spatial smoothness over geographic space. In relatively flat terrain, such an assumption may hold. However, in mountainous regions, two stations separated by a short geographic distance may exhibit significantly different temperature and humidity profiles due to elevation gradients, ridge barriers, valley inversions, and orographic effects. Conversely, distant valley stations may share similar atmospheric behavior despite being geographically far apart.

This phenomenon, referred to in this study as the “distance trap,” is illustrated in figure below.

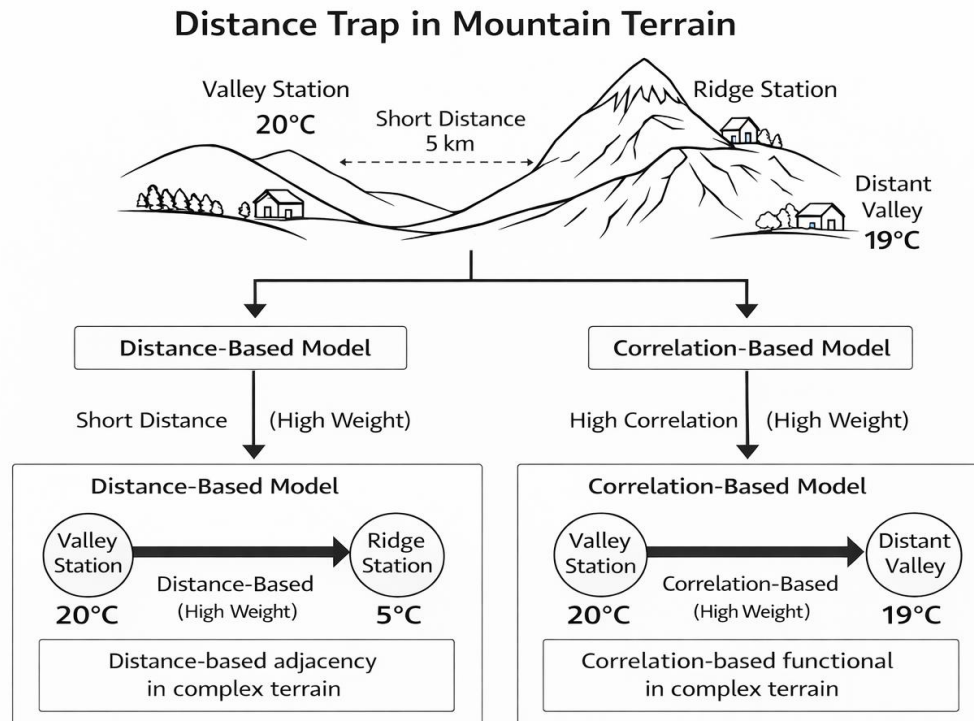


Figure 1.1 Conceptual Illustration of the “Distance Trap” in Mountain Terrain

This mismatch between geographic proximity and atmospheric similarity leads to incorrect information propagation within graph neural networks. As a result:

- Forecast accuracy decreases in high-relief zones
- Microclimate patterns are not adequately captured
- Crop recommendations based on these forecasts may become unreliable

The core problem addressed in this thesis is whether redefining spatial topology using functional connectivity measured through Pearson correlation of historical meteorological signals can improve forecasting performance and downstream crop recommendation accuracy in mountainous environments.

1.3 Research Objectives

The primary objective of this study is to improve weather forecasting and crop recommendation reliability in mountainous regions through correlation-adaptive graph modeling.

Specific objectives are:

1. To construct a correlation-based adjacency matrix using historical T2M time-series data from 320 weather stations across Nepal.
2. To implement a Spatio-Temporal Graph Convolutional Network (STGCN) using the constructed graph.
3. To compare forecasting performance between correlation-based and distance-based graph structures.
4. To evaluate improvements in crop suitability classification resulting from improved weather forecasts.
5. To analyze performance differences across terrain complexity levels (valley vs ridge, elevation bands).

1.4 Significance of the Study

This study contributes in three dimensions:

Methodological Contribution

It challenges the assumption that geographic proximity defines atmospheric similarity in mountainous terrain. It introduces a data-driven topology approach grounded in functional connectivity.

Technical Contribution

It integrates correlation-adaptive graph construction with STGCN forecasting for environmental modeling.

Practical Contribution

It enhances the reliability of crop recommendation systems in remote mountainous regions, potentially supporting agricultural resilience and informed decision-making.

CHAPTER 2

LITERATURE REVIEW

Spatio-temporal forecasting has evolved significantly with the introduction of Graph Neural Networks (GNNs). Graph Convolutional Networks (GCNs) [3] provide the foundational framework for modeling structured spatial dependencies. STGCN [4] and related architectures such as diffusion-based recurrent models [5] combine temporal convolution with graph convolution to capture both spatial and temporal dependencies for spatio-temporal prediction tasks. These models have been widely applied in traffic forecasting, air quality prediction, and environmental monitoring.

Most graph-based forecasting models define adjacency using geodesic distance or inverse distance weighting (IDW) schemes [1]. This approach assumes spatial continuity and smoothness. However, in mountainous regions, elevation differences and terrain structures break this assumption.

Climate network studies have shown that atmospheric systems exhibit teleconnections highlighting statistical relationships between geographically distant regions [6], [7]. These

relationships suggest that atmospheric similarity may not align strictly with geographic proximity. From a broader network theory perspective, graph topology significantly influences information propagation and representation learning in graph-based models [8], [9].

Crop recommendation systems rely on weather forecasts as primary inputs. Random Forest models [10] and rule-based systems such as FAO EcoCrop [11] determine crop suitability based on temperature, humidity, and rainfall thresholds. Errors in weather forecasting directly influence crop recommendations.

Despite advances in graph-based modeling, limited research has explored alternative topology definitions for mountainous meteorology. This study addresses that gap by evaluating correlation-based adjacency matrices within an STGCN framework.

CHAPTER 3

METHODOLOGY

3.1 Process Method

The methodology of this study consists of five major stages:

1. Data Collection
2. Pre-Processing
3. Data Splitting
4. Model Training
5. Classification-based Agricultural Evaluation

The overall workflow integrates correlation-adaptive graph construction with spatio-temporal forecasting and downstream crop suitability analysis.

3.1.1 Data Collection

The dataset consists of approximately 320 meteorological stations maintained by the Department of Hydrology and Meteorology (DHM), Nepal. Historical daily or hourly records spanning multiple years will be used for modeling and evaluation.

The selected meteorological variables include temperature at 2 meters (T2M), relative humidity (RH2M), precipitation (PRECTOT), surface pressure (PS), wind speed at 10 meters (WS10M), and altitude as a static terrain feature.

The detailed description of the variables used in this study is presented below.

Variable Name	Abbreviation	Type	Description	Role in Model
Temperature at 2m	T2M	Continuous	Air temperature measured at 2 meters above ground	Primary correlation signal + Forecast target
Relative Humidity	RH2M	Continuous	Percentage of	Forecast target

			moisture in air	
Precipitation	PRECTOT	Continuous	Total rainfall over interval	Forecast target
Surface Pressure	PS	Continuous	Atmospheric pressure at surface	Auxiliary predictor
Wind Speed at 10m	WS10M	Continuous	Horizontal wind velocity	Auxiliary predictor
Altitude	—	Static	Elevation above sea level	Terrain feature

Table 3.1: Meteorological Variables Used in the Study

After collecting the data, it is organized into a time-indexed format suitable for spatio-temporal modeling.

3.1.2 Pre-Processing

Pre-processing ensures data consistency and model stability. The following steps are performed:

- Handling missing values using interpolation techniques
- Outlier detection and removal
- Feature normalization
- Transformation into a station-wise time-series matrix

To construct the correlation-adaptive graph, a pivot table of Date $\times$ Station for T2M is created. Pearson correlation is computed between station pairs:

$$A_{ij}^{corr} = |Pearson(T_i, T_j)|$$

Similar correlation-based climate network approaches have been explored in large-scale atmospheric analysis [5].

Only the top 33% strongest correlations are retained to ensure sparsity and remove noise. This results in a functional connectivity graph representing statistical similarity rather than geographic distance.

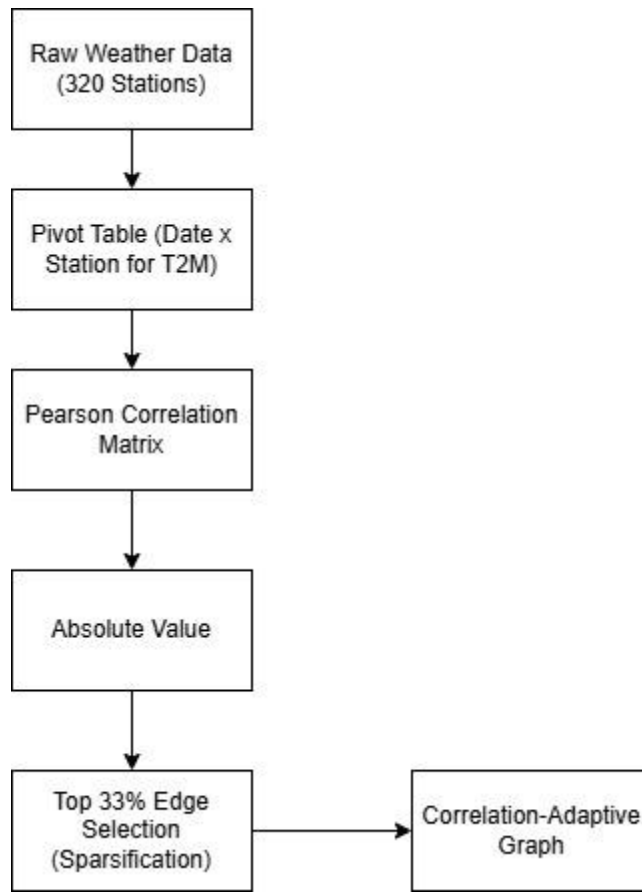


Figure 3.1: Workflow of correlation-adaptive weather graph construction from raw meteorological data to sparsified functional connectivity graph.

3.1.3 Data Splitting

A time-based split is applied to prevent data leakage:

- 70% Training
- 15% Validation
- 15% Testing

This ensures the model predicts future observations based on past data.

3.1.4 Model Training

The forecasting model implemented in this study is a Spatio-Temporal Graph Convolutional Network (STGCN) originally proposed for traffic forecasting applications [4]. The model integrates temporal convolution layers and graph convolution layers to capture both time-dependent and spatial dependencies across weather stations. The architecture consists of:

- Temporal gated convolution layers (kernel size=3)
- Chebyshev graph convolution layers based on spectral graph convolution theory [3] (K=3)
- A second spatio-temporal block for deeper feature extraction
- A fully connected output layer for multi-step forecasting

The model produces forecasts for temperature, humidity, and precipitation over a 7-day horizon.

The overall STGCN architecture used in this study is illustrated in following figure.

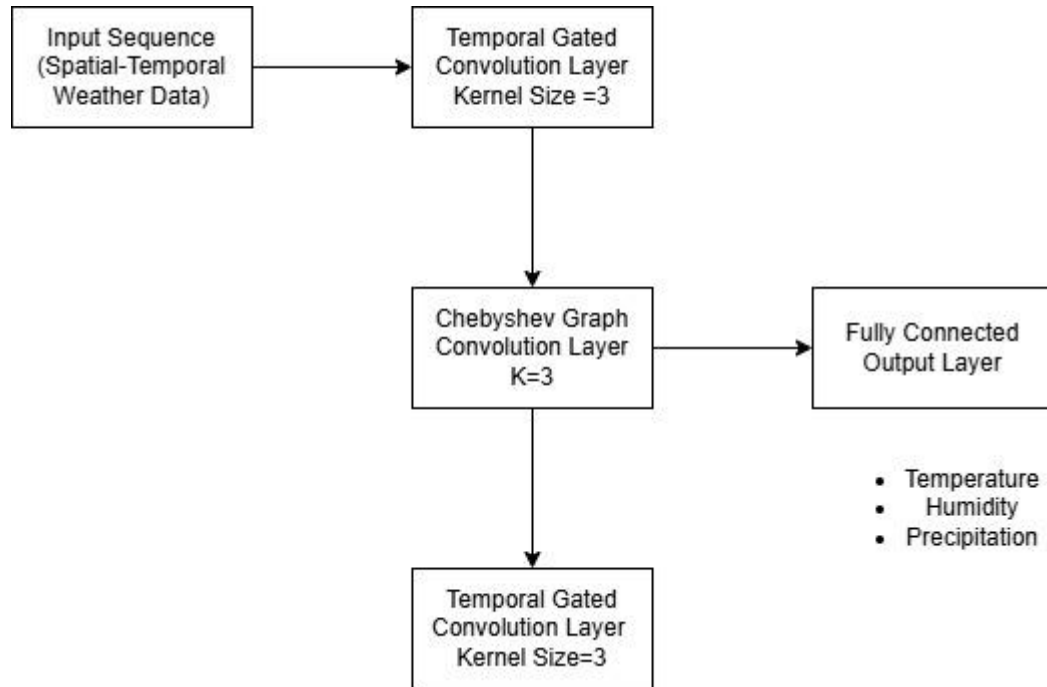


Figure 3.2 STGCN architecture overview

- Distance-based adjacency matrix
- Correlation-based adjacency matrix (proposed)

The configuration of the STGCN model is summarized in the table below.

Layer	Type	Parameters	Output Channels	Purpose
Input Layer	Feature Mapping	—	64	Projects input features into latent space
ST-Conv Block 1	Temporal Gated Conv + Chebyshev Graph Conv	Kernel=3, K=3	128	Captures local temporal and spatial patterns
ST-Conv Block 2	Temporal Gated Conv + Chebyshev Graph Conv	Kernel=3, K=3	256	Learns higher-level dependencies
Output Layer	Fully Connected	—	Forecast Horizon × Variables	Produces 7-day weather forecast

Table 3.2: Model Architecture Configuration

Forecast horizon: 7 days

Edge sparsity: Top 33% strongest correlations retained

The model is trained using backpropagation with mean squared error as the primary loss function. Hyperparameters are tuned using validation data.

Baselines include:

- Distance-based STGCN [4]
- LSTM without graph structure [12]
- Inverse Distance Weighting (IDW) interpolation [1]

3.1.5 Classification

The predicted weather outputs generated by the STGCN model are integrated into a crop suitability evaluation stage. Specifically, forecasted temperature, relative humidity, and precipitation values over the 7-day horizon are used as input features for determining crop compatibility under projected environmental conditions.

The crop recommendation module evaluates suitability based on climatic thresholds and compatibility analysis. For example, rice cultivation typically requires:

- Temperature $> 20^{\circ}\text{C}$
- Relative Humidity $> 80\%$

These threshold-based criteria are used to determine whether forecasted environmental conditions support a given crop. In addition to rule-based suitability assessment, a supervised learning approach such as a Random Forest classifier may be applied to model crop suitability using historical yield and climate records.

The integration of weather forecasting with the crop recommendation framework is illustrated in the figure below.

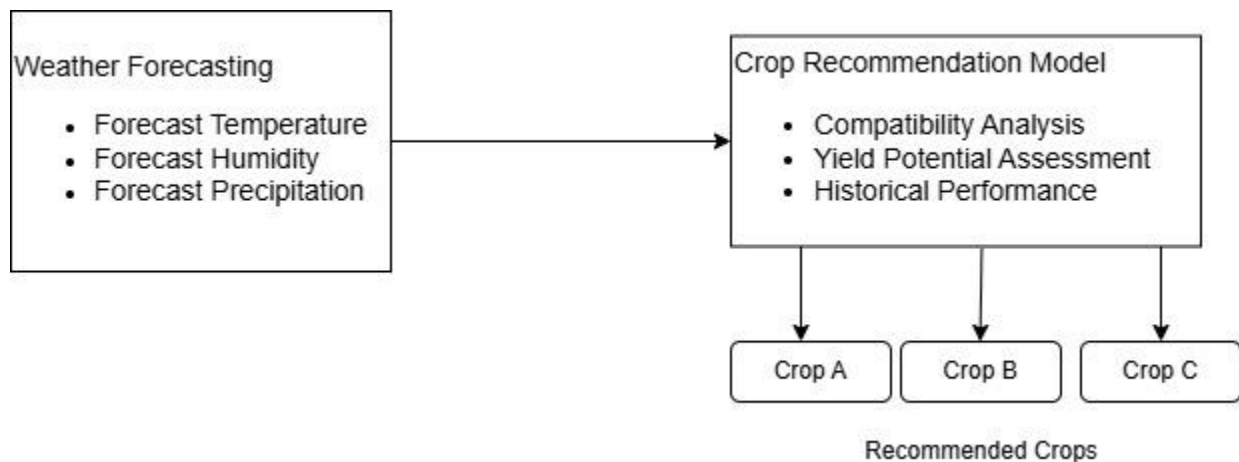


Figure 3.3 Integration of Weather Forecasting into Crop Recommendation

The final output of this stage is a ranked list of recommended crops based on forecasted environmental compatibility. By comparing recommendations derived from distance-based and correlation-based forecasting models, the study evaluates whether improvements in meteorological prediction translate into more reliable agricultural decision-making.

3.2 Validation and Verification

Model validation is conducted on the validation dataset during training. Final performance is assessed on the independent test dataset.

Verification includes:

- Comparative analysis between correlation-based and distance-based graphs
- Terrain-stratified evaluation (valley vs ridge, elevation bands)
- Sensitivity analysis on sparsification ratio

This ensures robustness of results across different geographic conditions.

3.3 Evaluation Metrics

Forecasting and classification performance are evaluated using quantitative metrics. These metrics are summarized in the following table.

Category	Metric	Formula	Purpose
Forecasting	RMSE	$\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$	Measures magnitude of prediction error
Forecasting	MAE	$\frac{1}{n} \sum y_i - \hat{y}_i $	Measures average absolute prediction error
Forecasting	R ²	$1 - \frac{SS_{res}}{SS_{tot}}$	Measures variance explained
Classification	Accuracy	$(TP + TN) / \text{Total}$	Overall correctness
Classification	Precision	$TP / (TP + FP)$	Reliability of positive recommendations
Classification	Recall	$TP / (TP + FN)$	Sensitivity to suitable crops
Classification	F1-score	$2(\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Balanced classification measure

Table 3.3: Forecasting and Classification Evaluation Metrics

3.4 Environment and Tools

The implementation environment includes:

- Python programming language
- PyTorch deep learning framework
- torch_geometric_temporal library
- Pandas and NumPy for preprocessing
- GPU-enabled computing environment

CHAPTER 4

EXPECTED OUTCOMES

The study is expected to demonstrate that correlation-based graph topology improves forecasting accuracy in mountainous terrain compared to distance-based topology.

The study hypothesizes that:

- Forecast error (RMSE, MAE) will decrease significantly in high-relief zones.
- Terrain-aware performance differences will be observable.
- Crop classification accuracy will improve when using correlation-based forecasts.

The outcome will provide a validated framework for weather-driven crop recommendation systems in mountainous regions.

CHAPTER 5

THESIS SCHEDULES

Month	Activity
1-2	Data preprocessing and graph construction
3	Baseline implementation
4-5	Model training and tuning
6	Comparative evaluation
7	Agricultural validation
8	Thesis writing and revision

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